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MILITARY STANDARD

WAVEGUIDE ASSEMBLIES, RIGID AND FLEXIBLE, SELECTION OF

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FSC 5985

DEPARTMENT OF DEFENSE
Washington, DC 20301

Waveguide Assemblies, Rigid, and Flexible, Selection of.

MIL-STD-1638A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ELEX 5043, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 General. This standard defines waveguide assemblies considered by the Department of Defense as standard for use in military equipment and applications.

1.2 Purpose. The purpose of this standard is to:

a. Provide the equipment designer with a list of waveguide assemblies considered standard for use in military applications.
b. Restrict and minimize the variety of waveguide assemblies for use in military applications in order to provide effective logistic support of equipment.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-287 - Waveguide Assemblies, Flexible, Twistable and Nontwistable, General Specification For.
MIL-W-3970 - Waveguide Assemblies, Rigid, General Specification For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are those commonly encountered in microwave engineering practice.

4. GENERAL REQUIREMENTS

4.1 Selection of waveguide assemblies. Waveguide assemblies to be used in military applications shall be selected from those listed in the tables herein.

4.2 Criteria for inclusion. The criteria for the selection of waveguide assemblies for inclusion in this standard are:

a. The waveguide assembly shall be considered by representatives of the Military Departments as the best available type for the current application.
b. Availability of the waveguide assembly shall be reasonably certain.
c. The waveguide assembly shall have an approved military specification.

4.3 Electrical and physical tolerances. Waveguide assemblies used in military applications shall be representative of manufactured lots possessing acceptable materials, physical and electrical characteristics, and shall in no manner degrade the operational characteristics of the equipment used in military applications.

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4.4 Detailed requirements. The detailed requirements for waveguide assemblies listed in this standard are covered by MIL-W-3970 or MIL-W-287 specification sheets, as applicable.

5. DETAILED REQUIREMENTS

Not applicable.

6. NOTES

Not applicable.

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TABLE I. Waveguide assemblies, rigid, straight, 90-degree twist.

Dash No. M3970/1-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.02	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.2	-	1.05	Cu	↓
004	1-073	53-002	59-007	7.05 - 10	-	1.05	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	Cu	
006	1-102	54-001	59-003	18 - 26.5	.11	1.06	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.06	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	Al	
009	1-071	53-004	59-009	7.05 - 10	-	1.05	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.02	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	Al	↓
013	3-006	54-003	59-005	26.5 - 40	.15	1.10	Cu	
014	1-065	55-002	60-002	5.85 - 8.2	-	1.05	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.11	1.06	Al	
017	1-157	70-002	70-002	7.0 - 11	-	1.06	Al	
018	1-073	59-007	59-007	7.05 - 10	-	1.05	Cu	
019	1-079	59-006	59-006	8.2 - 12.4	-	1.05	Cu	
020	1-090	53-006	59-002	12.4 - 18	.07	1.06	Al	
021	1-090	53-006 2/	53-006 2/	12.4 - 18	.07	1.06	Cu	↓
022	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
023	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
024	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
025	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
026	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
027	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.
2/ Except flange thickness is .25 in lieu of .125.

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TABLE II. Waveguide assemblies, rigid, 45-degree E-plane bend.

Dash No. M3970/5-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.04	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	-	1.05	Cu	↓
004	1-073	53-002	59-007	7.05 - 10	-	1.05	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.04	Cu	
006	1-102	54-001	59-003	18 - 26.5	.11	1.05	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.05	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.04	Al	
009	1-071	53-004	59-009	7.05 - 10	-	1.05	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.04	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	Al	↓
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	-	1.05	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.11	1.05	Al	
017	3-006 3-008	54-003	59-005	26.5 - 40	.15	1.1	Cu	30
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE III. Waveguide assemblies, rigid, 60-degree E-plane bend.

Dash No. M3970/6-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.02	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.2	-	1.05	Al	
004	1-073	53-002	59-007	7.05 - 10	-	1.04	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	Cu	
006	1-102	54-001	59-003	18 - 26.5	.11	1.05	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.05	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	Cu	
009	1-071	53-004	59-009	7.05 - 10	-	1.04	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.02	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	Al	
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	-	1.05	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.11	1.05	Al	
017	3-006 3-008	54-003	59-005	26.5 - 40	.15	1.1	Cu	30
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE IV. Waveguide assemblies, rigid, 90-degree E-plane bend.

Dash No. M3970/7-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.03	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.2	-	1.05	Cu	
004	1-073	53-002	59-007	7.05 - 10	-	1.05	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	Cu	
006	1-102	54-001	59-003	18 - 26.5	.11	1.06	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.08	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	Al	
009	1-071	53-004	59-009	7.05 - 10	-	1.05	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.03	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	Al	
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	Cu	
014	1-065	55-002	60-002	5.85 - 8.2	-	1.05	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.11	1.06	Al	
017	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---
023	1-001 1-002 1-161	76-07	76-07	.32 - .49	.10	1.05	Al	.25
024	1-003 1-004 1-162	76-06	76-06	.35 - .53	.10	1.05	Al	
025	1-005 1-006 1-163	76-05	76-05	.410 - .625	.10	1.05	Al	
026	1-007 1-008 1-164	76-04	76-04	.490 - .750	.10	1.05	Al	.25
027	1-009 1-010 1-165	76-03	76-03	.64 - .96	.10	1.05	Al	.50
028	1-011 1-012 1-166	76-02	76-02	.75 - 1.12	.10	1.05	Al	.50
029	1-013 1-014 1-167	76-01	76-01	.96 - 1.45	.10	1.05	Al	1.0

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE IV. Waveguide assemblies, rigid, 90-degree E-plane bend - Continued.

Dash No. M3970/7-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
030	1-018 1-019 1-168	52-002	52-002	1.12 - 1.70	.10	1.05	Al	1.5
031	1-029 1-030 1-170	52-006	52-006	1.70 - 2.60	.10	1.05	Al	3.0

1/ Cu = Copper alloy, Al = Aluminum alloy.

TABLE V. Waveguide assemblies, rigid, 45-degree E-plane mitered corner.

Dash No. M3970/8-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.08	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.08	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.07	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.07	Cu	
005	1-079	53-001	59-006	8.20 - 12.4	.06	1.08	Cu	
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.06	Cu	
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.06	Cu	
008	1-077	53-003	59-008	8.20 - 12.4	.06	1.08	Al	
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.08	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.08	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.08	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.12	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.07	Al	
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.06	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE VI. Waveguide assemblies, rigid, 60-degree E-plane mitered corner.

Dash No. M3970/9-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.07	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.09	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.10	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.08	Cu	
005	1-079	53-001	59-006	8.20 - 12.4	.06	1.09	Cu	
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.10	Cu	
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.10	Cu	
008	1-077	53-003	59-008	8.20 - 12.4	.06	1.09	Al	
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.08	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.07	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.09	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.09	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.15	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.10	Al	
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.09	Cu	
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.10	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

TABLE VII. Waveguide assemblies, rigid, 90-degree E-plane mitered corner.

Dash No. M3970/10-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.07	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.09	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.10	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.05	Cu	
005	1-079	53-001	59-006	8.20 - 12.40	.06	1.10	Cu	
006	1-102	54-001	59-003	18.0 - 26.50	.11	1.07	Cu	
007	1-106	54-001	59-003	18.0 - 26.50	.11	1.16	Cu	
008	1-077	53-003	59-008	8.20 - 12.40	.06	1.10	Al	
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.05	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.07	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.09	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.16	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.10	Al	
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.07	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE VIII. Waveguide assemblies, rigid, 45-degree H-plane bend.

Dash No. M3970/11-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.04	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.2	-	1.04	Cu	↓
004	1-073	53-002	59-007	7.05 - 10	-	1.05	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.04	Cu	
006	1-102	54-001	59-003	18 - 26.5	-	1.06	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.06	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.04	Al	
009	1-071	53-004	59-009	7.05 - 10	-	1.05	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.04	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.04	Al	↓
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	Cu	
014	1-065	55-002	60-002	5.85 - 8.2	-	1.05	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.15	1.06	Al	
017	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE IX. Waveguide assemblies, rigid, 60-degree H-plane bend.

Dash No. M3970/12-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.04	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.04	Cu	30
003	1-067	55-001	60-001	5.85 - 8.2	-	1.04	Cu	↓
004	1-073	53-002	59-007	7.05 - 10	-	1.05	Cu	
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	Cu	
006	1-102	54-001	59-003	18 - 26.5	.11	1.04	Cu	
007	1-106	54-001	59-003	18 - 26.5	.11	1.04	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	Al	↓
009	1-071	53-004	59-009	7.05 - 10	-	1.05	Al	30
010	1-041	56-002	61-001	2.6 - 3.95	-	1.04	Al	15
011	1-089	53-005	59-001	12.4 - 18	.07	1.04	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.04	Al	↓
013	3-006	54-003	59-005	26.5 - 40	.15	1.10	Cu	
014	1-065	55-002	60-002	5.85 - 8.2	-	1.04	Al	
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	Cu	
016	1-103	54-002	59-004	18 - 26.5	.11	1.04	Al	↓
017	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE X. Waveguide assemblies, rigid, 90-degree H-plane bend.

Dash No. M3970/13-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.6 - 3.95	-	1.05	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	-	1.03	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	-	1.07	Cu	
005	1-079	53-001	59-006	8.20 - 12.4	-	1.05	Cu	
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.05	Cu	
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.05	Cu	
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	Al	
009	1-071	53-004	59-009	7.05 - 10.0	-	1.07	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	-	1.05	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.04	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.05	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	-	1.03	Al	
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.04	Cu	
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.05	Cu	
017	1-018	52-002	52-002	1.12 - 1.70	-	1.05	Al	30
018	3-006	54-003	54-003	26.5 - 40	.15	1.1	Cu	---
019	3-006	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
020	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
021	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
022	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
023	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---
024	1-001	76-07	76-07	.32 - .49	.10	1.05	Al	.25
025	1-003	76-06	76-06	.35 - .53	.10	1.05	Al	
026	1-005	76-05	76-05	.410 - .625	.10	1.05	Al	
027	1-007	76-04	76-04	.490 - .750	.10	1.05	Al	.25
028	1-009	76-03	76-03	.64 - .96	.10	1.05	Al	.50
029	1-011	76-02	76-02	.75 - 1.12	.10	1.05	Al	.50
030	1-013	76-01	76-01	.96 - 1.45	.10	1.05	Al	1.0
031	1-018	52-002	52-002	1.12 - 1.70	.10	1.05	Al	1.5
032	1-029	52-006	52-006	1.70 - 2.60	.10	1.05	Al	3.0

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE XI. Waveguide assemblies, rigid, 45-degree H-plane mitered corner.

Dash No. M3970/14-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.03	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.05	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.06	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.02	Cu	↓
005	1-079	53-001	59-006	8.20 - 12.4	.06	1.08	Cu	↓
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.04	Cu	↓
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.04	Cu	↓
008	1-077	53-003	59-008	8.20 - 12.4	.06	1.08	Al	↓
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.02	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.03	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.08	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.05	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.08	Cu	↓
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.06	Al	↓
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.08	Cu	↓
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.04	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

TABLE XII. Waveguide assemblies, rigid, 60-degree H-plane mitered corner.

Dash No. M3970/15-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.10	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.07	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.08	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.06	Cu	↓
005	1-079	53-001	59-006	8.20 - 12.4	.06	1.07	Cu	↓
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.09	Cu	↓
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.09	Cu	↓
008	1-077	53-003	59-008	8.20 - 12.4	.06	1.07	Al	↓
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.06	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.10	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.08	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.07	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.14	Cu	↓
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.08	Al	↓
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.08	Cu	↓
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.09	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

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TABLE XIII. Waveguide assemblies, rigid, 90-degree H-plane mitered corner.

Dash No. M3970/16-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.09	Cu	15
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.08	Cu	30
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.08	Cu	↓
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.08	Cu	
005	1-079	53-001	59-006	8.20 - 12.40	.06	1.10	Cu	
006	1-102	54-001	59-003	18.0 - 26.50	.11	1.06	Cu	
007	1-106	54-001	59-003	18.0 - 26.50	.11	1.06	Cu	
008	1-077	53-003	59-008	8.20 - 12.40	.06	1.10	Al	
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.08	Al	30
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.09	Al	15
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	30
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.08	Al	↓
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.12	Cu	
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.08	Al	
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.10	Cu	
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.06	Al	30

1/ Cu = Copper alloy, Al = Aluminum alloy.

TABLE XIV. Waveguide assemblies, rigid, straight 45-degree twist.

Dash No. M3970/17	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
002	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
003	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
004	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
005	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
006	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy.

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TABLE XV. Waveguide assemblies, rigid, 30-degree E-plane bend.

Dash No. M3970/18-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
				GHz	dB			
001	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
002	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
003	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
004	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
005	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
006	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy.

TABLE XVI. Waveguide assemblies, rigid, 30-degree H-plane bend.

Dash No. M3970/19-	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material 1/	Pressurization $lb_f/in.^2$ gage
		Configuration B per MIL-W-3970/19		GHz	dB			
001	2/			1.25 - 1.35	-	1.05	Al	5
002	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	Cu	30
003	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	Cu	---
004	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	Cu	---
005	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	Cu	---
006	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	Cu	---
007	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	Cu	---

1/ Cu = Copper alloy, Al = Aluminum alloy.
2/ M85/1-018, except B dimension shall be 1.000 and D dimension 1.500.

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TABLE XVII. Waveguide assembly, rigid, 40-degree H-plane bend.

Dash No. M3970/20-	Waveguide M85/1-	Flange 1	Flange 2	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Material	Pressurization $lb_f/in.^2$ gage max
		M3970/20-		GHz	dB			
001	018 1/	001, figure 1		1.25 - 1.35	.05	1.05	Aluminum alloy	5

1/ Except B dimension shall be 1.000 and D dimension shall be 1.500.

TABLE XVIII. Waveguide assemblies, rigid, cast.

Part No. M3970/	Frequency range TE_{10} mode (GHz)	VSWR (max)	Mates with standard rigid waveguide M85/1	Material 1/	Pressurization $lb_f/in.^2$ gage (max)
45-degree E-plane bend, full corral and self-jigging					
21-001	8.2-12.4	1.08	077,078 178	Al	45
21-002	8.2-12.4	1.08	077,078 178	Al	45
90-degree E-plane bend, full corral					
22-001	3.95-5.85	1.05	053,054 174	Al	45
22-002	7.05-10.0		069,073	Cu	
22-003	7.05-10.0		071,072 177	Al	
22-004	7.05-10.0		069,073	Cu	
22-005	7.05-10.0		071,072 177	Al	
22-006	8.2-12.4		075,079	Cu	
22-007	8.2-12.4		077,078 178	Al	
22-008	8.2-12.4		075,079	Cu	
22-009	8.2-12.4		077,078 178	Al	

See footnotes at end of table.

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TABLE XVIII. Waveguide assemblies, rigid, cast - Continued.

Part No. M3970/	Frequency range TE_{10} mode (GHz)	VSWR (max)	Mates with standard rigid waveguide M85/1	Material 1/	Pressurization $lb_f/in.^2$ gage (max)
90-degree E-plane bend, full corral - Continued.					
22-010	8.2-12.4	1.05	075,079	Cu	45
22-011	8.2-12.4		077,078 178	Al	
22-012	12.4-18.0		087,089	Cu	
22-013	12.4-18.0		090,091 180	Al	
22-014	12.4-18.0		087,089	Cu	
22-015	12.4-18.0		090,091 180	Al	
22-016	26.5-40.0		007,008 2/	Cu	
22-017	26.5-40.0		009 2/	Al	
90-degree E-plane bend, miter and radius-back miter, full corral					
23-001	4.0-5.9	1.05	053,054 174	Al	20
23-002	4.0-5.9	1.05	051,055	Cu	20
23-003	8.2-12.4	1.05	077,078 178	Al	45
45-degree E-plane bend, reduced height, full corral					
24-001	8.2-12.4	1.08	---	Al	45
90-degree E-plane bend, reduced height, radius-back miter, full corral					
25-001	8.2-12.4	1.05	---	Al	45
45-degree H-plane bend, full corral and self-jigging					
26-001	8.2-12.4	1.05	077,078 178	Al	45
26-002	8.2-12.4	1.05	077,078 178	Al	45

See footnotes at end of table.

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TABLE XVIII. Waveguide assemblies, rigid, cast - Continued.

Part No. M3970/	Frequency range TE_{10} mode (GHz)	VSWR (max)	Mates with standard rigid waveguide M85/1	Material 1/	Pressurization $lb_f/in.^2$ gage (max) 2/
60-degree H-plane bend, full corral					
27-001	12.4-18.0	1.05	090,091 180	Al	45
90-degree H-plane bend, full corral					
28-001	8.2-12.4	1.05	077,078 178	Al	45
90-degree H-plane bend, miter and radius-back miter, full corral					
29-001	8.2-12.4	1.04	077,078 178	Al	45
29-002	8.2-12.4	1.04	077,078 178	Al	45
45-degree H-plane bend, reduced height, full corral					
30-001	8.2-12.4	1.05	---	Al	45
90-degree H-plane bend, reduced height, full corral					
31-001	8.2-12.4	1.05	---	Al	45

1/ Cu = Copper alloy, Al = Aluminum.
2/ M85/3.

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TABLE XIX. Waveguide assemblies, flexible, twistable, class 1.

Part number M287/3-1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
01-XXXX 02-XXXX 60-XXXX 3/	1.12-1.70	24.12	48.08	.005	1.10	017 018 017	58-007 58-008 58-007	58-007 58-008 58-009	Cu Al Cu	2
03-XXXX 04-XXXX 61-XXXX 3/	1.70-2.60	12.39	25.95	.01	1.10	031 029 031	58-009 58-010 58-009	58-009 58-010 58-009	Cu Al Cu	4
05-XXXX 06-XXXX 62-XXXX 3/	2.20-3.30	11.14	19.64	.015	1.10	037 035 037	58-011 58-012 58-011	58-011 58-012 58-011	Cu Al Cu	5
07-XXXX 08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 63-XXXX 3/	2.60-3.95	6.88	14.38	.02	1.10	043 041 043 041 043 041 043 041 043	64-001 64-002 56-001 56-002 61-002 61-001 61-002 61-001 61-002	64-001 64-002 56-001 56-002 56-001 56-002 61-002 61-001 56-001	Cu Al Cu Al Cu Al Cu Al Cu	15
15-XXXX 16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 64-XXXX 3/	3.95-5.85	4.94	12.04	.05	1.10	055 053 055 053 055 053 055 053 055	63-001 63-005 57-002 57-001 62-002 62-001 62-002 62-001 63-001	63-001 63-005 57-002 57-001 57-002 57-001 62-002 62-001 63-001	Cu Al Cu Al Cu Al Cu Al Cu	30
23-XXXX 24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 65-XXXX 3/	5.85-8.20	3.88	9.25	.08	1.10	067 065 067 065 067 065 067 065 067	63-002 63-006 55-001 55-002 60-001 60-002 60-001 60-002 63-002	63-002 63-006 55-001 55-002 55-001 55-002 60-001 60-002 63-002	Cu Al Cu Al Cu Al Cu Al Cu	30
31-XXXX 32-XXXX 33-XXXX 34-XXXX 35-XXXX 36-XXXX 37-XXXX 38-XXXX 66-XXXX 3/	7.05-10.00	3.57	8.38	.08	1.15	073 071 073 071 073 071 073 071 073	63-003 63-007 53-002 53-004 59-007 59-009 59-007 59-009 63-003	63-003 63-007 53-002 53-004 53-002 53-004 59-007 59-007 63-003	Cu Al Cu Al Cu Al Cu Al Cu	30

See footnotes at end of table.

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TABLE XIX. Waveguide assemblies, flexible, twistable, class 1 - Continued.

Part number M287/3- 1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
39-XXXX	8.20-12.40	3.38	7.51	.1	1.15	079	63-004	63-004	Cu	30
40-XXXX						077	63-008	63-008	Al	
41-XXXX						079	53-001	53-001	Cu	
42-XXXX						077	53-003	53-003	Al	
43-XXXX						079	59-006	53-001	Cu	
44-XXXX						077	59-008	53-003	Al	
45-XXXX						079	59-006	59-006	Cu	
46-XXXX						077	59-008	59-008	Al	
67-XXXX 3/						079	63-004	63-004	Cu	
47-XXXX	12.40-18.00	2.75	5.75	.25	1.20	089	53-005	53-005	Cu	30
48-XXXX						090	53-006	53-006	Al	
49-XXXX						089	59-001	53-005	Cu	
50-XXXX						090	59-002	53-006	Al	
51-XXXX						089	59-001	59-001	Cu	
52-XXXX						090	59-002	59-002	Al	
53-XXXX						089	53-005	53-005	Cu	
68-XXXX 3/						089	53-005	53-005	Cu	
54-XXXX	18.0-26.5	2.33	4.5	.35	1.20	102	54-001	54-001	Cu	30
55-XXXX						103	54-002	54-002	Al	
56-XXXX						102	59-003	54-001	Cu	
57-XXXX						103	59-004	54-002	Al	
58-XXXX						102	59-003	59-003	Cu	
59-XXXX						103	59-004	59-004	Al	
69-XXXX 3/						102	54-001	54-001	Cu	

1/ The last four digits shall indicate the nominal relaxed length.
2/ Cu = Copper alloy, Al = Aluminum alloy.
3/ The assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5. All other characteristics are as listed.

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TABLE XX. Waveguide assemblies, flexible, nontwistable, class 2.

Part number M287/4-1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch) E plane	Minimum centerline bending radius (inch) H plane	Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges Flange 1 M3922/	Flanges Flange 2 M3922/	Assembly material 2/	Pressurization $lb_f/in.^2$ gage
01-XXXX 02-XXXX 03-XXXX 64-XXXX 3/	1.12-1.70	13.02	27.07	.01	1.05	017 018 017 017	58-007 58-008 58-007 58-007	58-007 58-008 58-007 58-007	Cu Al Cu Cu	15
04-XXXX 05-XXXX 65-XXXX 3/	1.70-2.60	9.51	14.00	.015	1.07	031 029 031	58-009 58-010 58-009	58-009 58-010 58-009	Cu Al Cu	20
06-XXXX 07-XXXX 66-XXXX 3/	2.20-3.30	5.00	9.75	.018	1.09	037 035 037	58-011 58-012 58-011	58-011 58-012 58-011	Cu Al Cu	25
08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 15-XXXX 67-XXXX 3/	2.60-3.95	4.50	6.38	.02	1.10	043 041 043 041 043 041 043 041 043	64-001 64-002 56-001 56-002 61-002 61-001 61-002 61-001 64-001	64-001 64-002 56-001 56-002 56-001 56-002 61-002 61-001 64-001	Cu Al Cu Al Cu Al Cu Al Cu	30
16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 23-XXXX 68-XXXX 3/	3.95-5.85	3.00	5.50	.03	1.10	055 053 055 053 055 053 055 053 055	63-001 63-005 57-002 57-001 62-002 62-001 62-002 62-001 63-001	63-001 63-005 57-002 57-001 57-002 57-001 62-002 62-001 63-001	Cu Al Cu Al Cu Al Cu Al Cu	30
24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 31-XXXX 32-XXXX 69-XXXX 3/	5.85-8.20	2.50	5.00	.05	1.10	067 065 067 065 067 065 067 065 065 067	63-002 63-006 55-001 55-002 60-001 60-002 60-001 60-002 63-006 63-002	63-002 63-006 55-001 55-002 55-001 55-002 60-001 60-002 63-006 63-002	Cu Al Cu Al Cu Al Cu Al Al Cu	30

See footnotes at end of table.

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TABLE XX. Waveguide assemblies, flexible, nontwistable, class 2 - Continued.

Part number M287/4-1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
33-XXXX	7.05-10.0	1.75	3.49	.06	1.10	073	63-003	63-003	Cu	45
34-XXXX						071	63-007	63-007	Al	
35-XXXX						073	53-002	53-002	Cu	
36-XXXX						071	53-004	53-004	Al	
37-XXXX						073	59-007	53-002	Cu	
38-XXXX						071	59-009	53-004	Al	
39-XXXX						073	59-007	59-007	Cu	
40-XXXX						071	59-009	59-009	Al	
70-XXXX 3/						073	63-003	63-003	Cu	
41-XXXX	7.0-11.0	1.44	2.50	.08	1.10	155	70-001	70-001	Cu	50
42-XXXX						157	70-002	70-002	Al	
71-XXXX 3/						155	70-001	70-001	Cu	
43-XXXX	8.2-12.4	1.38	3.00	.1	1.10	079	63-004	63-004	Cu	60
44-XXXX						077	63-008	63-008	Al	
45-XXXX						079	53-001	53-001	Cu	
46-XXXX						077	53-003	53-003	Al	
47-XXXX						079	59-006	53-001	Cu	
48-XXXX						077	59-008	53-003	Al	
49-XXXX						079	59-006	59-006	Cu	
50-XXXX						077	59-008	59-008	Al	
72-XXXX 3/						079	63-004	63-004	Cu	
51-XXXX	12.4-18.0	1.25	2.00	.25	1.13	089	53-005	53-005	Cu	60
52-XXXX						090	53-006	53-006	Al	
53-XXXX						089	59-001	53-005	Cu	
54-XXXX						090	59-002	53-006	Al	
55-XXXX						089	59-001	59-001	Cu	
56-XXXX						090	59-002	59-002	Al	
57-XXXX						090	53-006	53-006	Al	
73-XXXX 3/						089	53-005	53-005	Cu	
58-XXXX	18.0-26.5	1.13	1.25	.50	1.15	102	54-001	54-001	Cu	60
59-XXXX						103	54-002	54-002	Al	
60-XXXX						102	59-003	54-001	Cu	
61-XXXX						103	59-004	54-002	Al	
62-XXXX						102	59-003	59-003	Cu	
63-XXXX						103	59-004	59-004	Al	
74-XXXX 3/						102	54-001	54-001	Cu	

1/ The last four digits shall indicate the nominal relaxed length.
2/ Cu = Copper alloy, Al = Aluminum alloy.
3/ The assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5. All other characteristics are as listed.

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TABLE XXI. Waveguide assemblies, extra-flexible, twistable, class 3.

Part number M287/5-1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
01-XXXX 02-XXXX 60-XXXX 3/	1.12-1.70	13.02	27.07	.025	1.10	017 018 017	58-007 58-008 57-007	58-007 58-008 57-007	Cu Al Cu	2
03-XXXX 04-XXXX 61-XXXX 3/	1.70-2.60	9.51	14.00	.035	1.10	031 029 031	58-009 58-010 58-009	58-009 58-010 58-009	Cu Al Cu	20
05-XXXX 06-XXXX 62-XXXX 3/	2.20-3.30	6.00	8.00	.05	1.10	037 035 037	58-011 58-012 58-011	58-011 58-012 58-011	Cu Al Cu	30
07-XXXX 08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 15-XXXX 63-XXXX 3/	2.60-3.95	4.50	6.38	.06	1.10	043 041 043 041 043 041 043 041 041 043	64-001 64-002 56-001 56-002 61-002 61-001 61-002 61-001 52-032 58-011	64-001 64-002 56-001 56-002 56-001 56-002 61-002 61-001 52-032 58-011	Cu Al Cu Al Cu Al Cu Al Al Cu	30
16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 23-XXXX 64-XXXX 3/	3.95-5.85	3.00	4.75	.07	1.10	055 053 055 053 055 053 055 053 055	63-001 63-005 57-002 57-001 62-002 62-001 62-002 62-001 63-001	63-001 63-005 57-002 57-001 57-002 57-001 62-002 62-001 63-001	Cu Al Cu Al Cu Al Cu Al Cu	45
24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 31-XXXX 65-XXXX 3/	5.85-8.20	2.50	4.75	.08	1.12	067 065 067 065 067 065 067 065 067	63-002 63-006 55-001 55-002 60-001 60-002 60-001 60-002 63-002	63-002 63-006 55-001 55-002 55-001 55-002 60-001 60-002 63-002	Cu Al Cu Al Cu Al Cu Al Cu	45
32-XXXX 33-XXXX 34-XXXX 35-XXXX 36-XXXX 37-XXXX 38-XXXX 39-XXXX 66-XXXX 3/	7.05-10.0	2.00	4.00	.08	1.15	073 071 073 071 073 071 073 071 073	63-003 63-007 53-002 53-004 59-007 59-009 59-007 59-009 63-003	63-003 63-007 53-002 53-004 53-002 53-004 59-007 59-009 63-003	Cu Al Cu Al Cu Al Cu Al Cu	45

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TABLE XXI. Waveguide assemblies, extra-flexible, twistable, class 3 - Continued.

Part number M287/5- 1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M85/1-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
40-XXXX 41-XXXX 42-XXXX 43-XXXX 44-XXXX 45-XXXX 46-XXXX 47-XXXX 67-XXXX 3/	8.2-12.4	1.38	2.25	.13	1.15	079 077 079 077 079 077 079 077 079	63-004 63-008 53-001 53-003 59-006 59-008 59-006 59-008 63-004	63-004 63-008 53-001 53-003 53-001 53-003 59-006 59-008 63-004	Cu Al Cu Al Cu Al Cu Al Cu	60
48-XXXX 49-XXXX 50-XXXX 51-XXXX 52-XXXX 53-XXXX 68-XXXX 3/	12.4-18.0	1.25	1.75	.3	1.20	089 090 089 090 089 090 089	53-005 53-006 59-001 59-002 59-001 59-002 53-005	53-005 53-006 53-005 53-006 59-001 59-002 53-005	Cu Al Cu Al Cu Al Cu	60
54-XXXX 55-XXXX 56-XXXX 57-XXXX 58-XXXX 59-XXXX 69-XXXX 3/	18.0-26.5	1.12	1.25	.35	1.20	102 103 102 103 102 103 102	54-001 54-002 59-003 59-004 59-003 59-004 54-001	54-001 54-002 54-001 54-002 59-003 59-004 54-001	Cu Al Cu Al Cu Al Cu	60

1/ The last four digits shall indicate the nominal relaxed length.
2/ Cu = Copper alloy, Al = Aluminum alloy.
3/ The assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5. All other characteristics are as listed.

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TABLE XXII. Waveguide assemblies, flexible, twistable, class 5.

Part number M287/6 1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M23351/4-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M39000/	Flange 2 M39000/		
01-XXXX	3.5-8.2	3.5	7.0	.10	1.38	029	3-058	3-032	A1	30
02-XXXX	4.75-11.0	3.0	6.0	.15	1.38	033	3-070	3-044	A1	35
03-XXXX	7.5-18.0	2.0	4.0	.20	1.38	037	3-082	3-074	A1	45

1/ The last four digits shall indicate the nominal relaxed length.
2/ Al=Aluminum alloy.

TABLE XXIII. Waveguide assemblies, flexible-twistable, class 7.

Part number M287/7 1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide M23351/4-	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M39000/	Flange 2 M39000/		
01-XXXX	3.5-8.2	3.00	6.00	.10	1.38	029	3-058	3-032	A1	30
02-XXXX	4.75-11.0	1.75	3.50	.15	1.38	033	3-070	3-044	A1	35
03-XXXX	7.5-18.0	1.625	3.25	.20	1.38	037	3-082	3-074	A1	45

1/ The last four digits shall indicate the nominal relaxed length.
2/ Al=Aluminum alloy.

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TABLE XXIV. Waveguide assemblies, millimeter, flexible, nontwistable, class B.

Part number M287/8 1/	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	VSWR	Mates with standard rigid waveguide MB5/3	Flanges		Assembly material 2/	Pressurization $lb_f/in.^2$ gage
		E plane	H plane				Flange 1 M3922/	Flange 2 M3922/		
01-XXXX	26.5-40.0	26.62	53.2	.05	1.30	006	59-005	54-003	Cu	60
02-XXXX	33.0-50.0	13.75	27.0	.05	1.10	010	67-006	67-006	Cu	15
03-XXXX	40.0-60.0	9.18	18.58	.08	1.10	014	67-007	67-007	Cu	15
04-XXXX	50.0-75.0	7.64	15.28	.11	1.10	017	67-008	67-008	Cu	15
05-XXXX	60.0-90.0	6.88	13.75	.20	1.12	020	67-009	67-009	Cu	20
06-XXXX	75.0-110	6.88	13.75	.25	1.15	023	67-010	67-010	Cu	20

1/ The last four digits shall indicate the nominal relaxed length.
2/ Cu=Copper alloy.

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TABLE XXV. Cross-reference of AN nomenclature to part number.

Type designation	Part number M3970/	Type designation	Part number M3970/
UG-712/U	13-001	UG-760/U	8-001
UG-713/U	13-002	UG-761/U	8-002
UG-714/U	13-003	UG-762/U	8-003
UG-715/U	13-004	UG-763/U	8-004
UG-716/U	13-005	UG-764/U	8-005
UG-717/U	13-006	UG-765/U	8-006
UG-718/U	13-007	UG-766/U	8-007
UG-719/U	13-008	UG-767/U	8-008
UG-720/U	13-009	UG-768/U	8-009
UG-721/U	13-010	UG-769/U	8-010
UG-722/U	13-011	UG-770/U	8-011
UG-723/U	13-012	UG-771/U	8-012
UG-724/U	13-013	UG-772/U	8-013
UG-725/U	13-014	UG-773/U	8-014
UG-726/U	13-015	UG-774/U	8-015
UG-727/U	13-016	UG-775/U	8-016
UG-728/U	5-001	UG-776/U	11-001
UG-729/U	5-002	UG-777/U	11-002
UG-730/U	5-003	UG-778/U	11-003
UG-731/U	5-004	UG-779/U	11-004
UG-732/U	5-005	UG-780/U	11-005
UG-733/U	5-006	UG-781/U	11-006
UG-734/U	5-007	UG-782/U	11-007
UG-735/U	5-008	UG-783/U	11-008
UG-736/U	5-009	UG-784/U	11-009
UG-737/U	5-010	UG-785/U	11-010
UG-738/U	5-011	UG-786/U	11-011
UG-739/U	5-012	UG-787/U	11-012
UG-740/U	5-013	UG-788/U	11-013
UG-741/U	5-014	UG-789/U	11-014
UG-742/U	5-015	UG-790/U	11-015
UG-743/U	5-016	UG-791/U	11-016
UG-744/U	9-001	UG-792/U	14-001
UG-745/U	9-002	UG-793/U	14-002
UG-746/U	9-003	UG-794/U	14-003
UG-747/U	9-004	UG-795/U	14-004
UG-748/U	9-005	UG-796/U	14-005
UG-749/U	9-006	UG-797/U	14-006
UG-750/U	9-007	UG-798/U	14-007
UG-751/U	9-008	UG-799/U	14-008
UG-752/U	9-009	UG-800/U	14-009
UG-753/U	9-010	UG-801/U	14-010
UG-754/U	9-011	UG-802/U	14-011
UG-755/U	9-012	UG-803/U	14-012
UG-756/U	9-013	UG-804/U	14-013
UG-757/U	9-014	UG-805/U	14-014
UG-758/U	9-015	UG-806/U	14-015
UG-759/U	9-016	UG-807/U	14-016

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TABLE XXV. Cross-reference of AN nomenclature to part number - Continued.

Type designation	Part number M3970/	Type designation	Part number M3970/
UG-808/U	12-001	UG-860/U	7-001
UG-809/U	12-002	UG-861/U	7-002
UG-810/U	12-003	UG-862/U	7-003
UG-811/U	12-004	UG-863/U	7-004
UG-812/U	12-005	UG-864/U	7-005
UG-813/U	12-006	UG-865/U	7-006
UG-814/U	12-007	UG-866/U	7-007
UG-815/U	12-008	UG-867/U	7-008
UG-816/U	12-009	UG-868/U	7-009
UG-817/U	12-010	UG-869/U	7-010
UG-818/U	12-011	UG-870/U	7-011
UG-819/U	12-012	UG-871/U	7-012
UG-820/U	12-013	UG-872/U	7-013
UG-821/U	12-014	UG-873/U	7-014
UG-822/U	12-015	UG-874/U	7-015
UG-823/U	12-016	UG-875/U	7-016
UG-824/U	6-001	UG-877/U	15-001
UG-825/U	6-002	UG-878/U	15-002
UG-826/U	6-003	UG-879/U	15-003
UG-827/U	6-004	UG-880/U	15-004
UG-828/U	6-005	UG-881/U	15-005
UG-829/U	6-006	UG-882/U	15-006
UG-830/U	6-007	UG-883/U	15-007
UG-831/U	6-008	UG-884/U	15-008
UG-832/U	6-009	UG-885/U	15-009
UG-833/U	6-010	UG-886/U	15-010
UG-834/U	6-011	UG-887/U	15-011
UG-835/U	6-012	UG-888/U	15-012
UG-836/U	6-013	UG-889/U	15-013
UG-837/U	6-014	UG-890/U	15-014
UG-838/U	6-015	UG-891/U	15-015
UG-839/U	6-016	UG-892/U	15-016
UG-844/U	16-001	UG-893/U	10-001
UG-845/U	16-002	UG-894/U	10-002
UG-846/U	16-003	UG-895/U	10-003
UG-847/U	16-004	UG-896/U	10-004
UG-848/U	16-005	UG-897/U	10-005
UG-849/U	16-006	UG-898/U	10-006
UG-850/U	16-007	UG-899/U	10-007
UG-851/U	16-008	UG-900/U	10-008
UG-852/U	16-009	UG-901/U	10-009
UG-853/U	16-010	UG-902/U	10-010
UG-854/U	16-011	UG-903/U	10-011
UG-855/U	16-012	UG-904/U	10-012
UG-856/U	16-013	UG-905/U	10-013
UG-857/U	16-014	UG-906/U	10-014
UG-858/U	16-015	UG-907/U	10-015
UG-859/U	16-016	UG-908/U	10-016

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Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - MI, AV,
Navy - OS
Air Force - 17, 99, 85
DLA - ES

User activities:
Army - AR
Navy - MC, CG, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
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DOCUMENT IDENTIFIER AND TITLE MIL-STD-1638A - Waveguide Assemblies, Rigid and Flexible, Selection of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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